



Rollback Speech: Smart Feedback Prompts for Lost Utterances in Unstable Online Calls

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Abstract

We present a prototype system designed to enhance the user experience of online conferencing in the presence of poor network conditions. When connection issues cause parts of a speaker's audio to be lost, the system uses dual automatic speech recognition (ASR) outputs—one local and one remote—to detect which utterances were not received. Upon reconnection, the speaker is provided with extracted keywords representing the information that did not reach the listener, enabling efficient and targeted repetition without full redundancy. This helps maintain conversational flow and reduces frustration during virtual communication. The system comprises real-time transcription, network degradation simulation, discrepancy detection, and lightweight keyword extraction in a modular architecture. Our “Show and Tell” demonstration allows participants to experience how the system augments communication continuity, even under conditions of packet loss or delayed transmission.

Index Terms: speech recognition, human-computer interaction, computational paralinguistics

1. Introduction & General Description

Despite the convenience of online call software, these tools still suffer from persistent usability issues, particularly in the face of unstable network conditions. One particularly frustrating scenario arises when audio is cut off due to packet loss, fluctuating bandwidth, or spotty Wi-Fi and mobile reception [1, 2]. These disruptions can interrupt the natural flow of conversation and diminish comprehension—especially when speakers must repeat themselves without knowing exactly what the listener missed. This lack of feedback can lead to confusion, wasted effort, and a general “sense of disconnect” among participants [3].

To address this problem, we developed an early-stage prototype that intelligently enhances online call experiences by detecting when speech has been lost due to connection failures. Our system leverages Automatic Speech Recognition (ASR) on both sides of a call to monitor what was said and what was received in real time. When audio fails to transmit during a connection drop, the system identifies the discrepancy and generates a set of keywords to summarize the missing portion. These are shown to the speaker upon reconnection, enabling them to efficiently reiterate only the relevant points, rather than restating everything.

This approach fills a unique gap in the current landscape of conferencing tools. Solutions such as MeetDot [4] or translation-enhanced captioning systems [5] prioritize accessibility and comprehension but generally assume reliable, uninterrupted links. Similarly, captioning systems for education [6], and speech processing techniques for noisy environments or overlapping speech [7], are not designed to handle ad-hoc

transmission losses. Even comprehension enhancement strategies using artificial delays during collaboration [8] fall short of tackling the intermittent, real-time packet loss typical of home connections.

Our prototype targets cases where network disruptions cause speech loss. By comparing ASR transcripts from both speaker and listener clients, the system identifies missed utterances and prompts the speaker to rephrase only the lost content. This preserves conversational flow and reduces redundancy during critical exchanges like meetings or presentations. In our demo, a simulated two-client call includes a brief disconnection. Upon reconnection, the speaker sees keyword prompts summarizing what was not received—demonstrating a practical, user-focused solution for restoring continuity in unstable network conditions.

2. Theoretical Background

Conventional online conferencing systems often fail to handle transient network outages gracefully, leading to unintelligible speech or lost message segments. This creates significant communication bottlenecks, particularly when speakers must repeat entire portions of their message due to a short-lived connection loss. Our goal is to improve the subjective user experience in such situations by identifying and recovering only the most relevant speech fragments missed during disconnection.

The key objective of our system is to detect periods of network instability, compare what the speaker said versus what was received on the other end, and extract semantically meaningful keywords from the untransmitted segment. This allows the speaker to selectively reiterate only the missing information, maintaining conversational context and reducing frustration. We implemented this using a modular pipeline:

- **Speech Capture and Transcription:** Both client and server components use automatic speech recognition (ASR) to transcribe speech in real time.
- **Network Simulation:** To evaluate robustness, an artificial network degradation module simulates bandwidth drops and introduces packet loss.
- **Mismatch Detection:** Upon network recovery, the transcriptions from sender and receiver are compared to identify discrepancies.
- **Keyword Extraction:** From the unreceived segment, a lightweight keyword extraction algorithm identifies and displays essential terms back to the speaker to prompt summarization.

Compared to existing approaches, which primarily focus on either live captioning or improving bandwidth efficiency, our solution is unique in that it handles the aftermath of speech loss

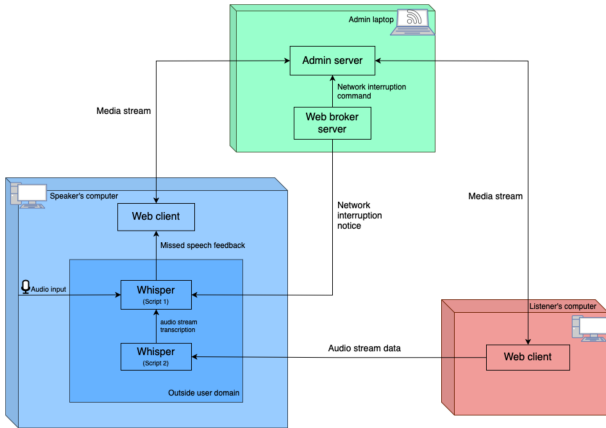


Figure 1: Block diagram of the demonstration system showing data flow and module interaction.

by offering real-time feedback to the speaker. Previous studies neglect this step, assuming stable transmission or relying purely on receiver-side retransmission strategies [9].

The proposed system is an early research prototype, but it opens the door to more nuanced user experience enhancements in VoIP applications, particularly for regions with unreliable infrastructure.

3. Detailed Workflow

To demonstrate the feasibility of our approach, we implemented a web-based prototype that mimics real-world online communication scenarios under unstable network conditions. The system enables one-way or two-way speech between participants, leveraging local transcription and network-aware logic to detect and handle disruptions.

3.1. System Architecture

The system consists of two primary client nodes—representing two users in a call—and a central server component that manages network simulation and message comparison. Below is a conceptual overview of the core workflow (see also Figure 1):

1. **Speech Capture:** The local client records and transcribes the speaker's utterances using a real-time automatic speech recognition (ASR) engine.
2. **Audio Transmission:** Audio data is transmitted to the receiving client via a WebSocket connection.
3. **Network Simulation:** Controlled packet loss and latency are introduced to simulate poor connectivity using an artificial degradation module.
4. **Discrepancy Detection:** When a disconnection occurs, transmitted and received transcripts are logged. Once the connection is restored, the system compares the complete local transcription with the (incomplete) received transcription.
5. **Keyword Extraction:** From the untransmitted portion, crucial keywords are extracted using a term frequency-based method and displayed to the speaker as feedback.
6. **Speaker Prompt:** The speaker sees the extracted keywords and is prompted to selectively rephrase the relevant part without repeating everything. This facilitates smoother communication recovery.

3.2. Implementation Details

Hardware and Environment: The current prototype operates on consumer-grade laptops as an interface, and uses a GPU server for the ASR via a network connection. The interface itself requires a standard web browser and provides a connection to its peer via WebRTC and WebSocket protocols. **Automatic Speech Recognition:** The demonstrated system uses the Whisper ASR system on our own servers for privacy and latency reasons [10]. **Keyword Algorithm:** A simple Python NLTK based keyword extraction approach was implemented to surface the most relevant words in the missing sequence. This way, the speaker can be provided with meaningful keywords while not having to read an entire sentence. **Software Stack:** The front-end system is implemented in Vue.JS and Node.JS. The back-end is written in python. Apart from the ASR component, the software used was created entirely for research purposes.

During the *Show and Tell* demonstration, two people will speak via the system across a table. A brief disconnection is initiated to simulate degraded connection. Upon reconnection, the visitor will see a feedback panel showing extracted keywords—showcasing how important speech elements can be preserved despite connectivity issues. This workflow offers a new user-centric approach to sustaining context in digital conversations.

4. References

- [1] J. Wang, "Chitchat: Making video chat robust to packet loss," Ph.D. dissertation, Massachusetts Institute of Technology, 2010.
- [2] C. Yu, Y. Xu, B. Liu, and Y. Liu, "“can you see me now?” a measurement study of mobile video calls," in *IEEE INFOCOM 2014-IEEE Conference on Computer Communications*. IEEE, 2014, pp. 1456–1464.
- [3] R. Bergmann, S. Rintel, N. Baym, A. Sarkar, D. Borowiec, P. Wong, and A. Sellen, "Meeting (the) pandemic: Videoconferencing fatigue and evolving tensions of sociality in enterprise video meetings during covid-19," *Computer Supported Cooperative Work (CSCW)*, vol. 32, no. 2, pp. 347–383, 2023.
- [4] A. Arkhangorodsky, C. Chu, S. Fang, Y. Huang, D. Jiang, A. Nagesh, B. Zhang, and K. Knight, "Meetdot: Videoconferencing with live translation captions," *arXiv preprint arXiv:2109.09577*, 2021.
- [5] S. Polepaka, V. P. Kumar, S. U. Chandra, H. N. S. Krishna, and G. Thakur, "Automated caption generation for video call with language translation," in *E3S Web of Conferences*, vol. 430. EDP Sciences, 2023, p. 01025.
- [6] R. Ranchal, T. Taber-Doughty, Y. Guo, K. Bain, H. Martin, J. P. Robinson, and B. S. Duerstock, "Using speech recognition for real-time captioning and lecture transcription in the classroom," *IEEE Transactions on Learning Technologies*, vol. 6, no. 4, pp. 299–311, 2013.
- [7] M. Yousefi, *Deep learning based methods for detection, separation, and recognition of overlapping speech*. The University of Texas at Dallas, 2021.
- [8] N. Yamashita, A. Echenique, T. Ishida, and A. Hautasaari, "Lost in transmittance: how transmission lag enhances and deteriorates multilingual collaboration," in *Proceedings of the 2013 conference on Computer supported cooperative work*, 2013, pp. 923–934.
- [9] D. Zhang, J. Liang, and I. Singh, "End-to-end distortion estimation for h. 264 with unconstrained intra prediction," in *2012 19th IEEE International Conference on Image Processing*. IEEE, 2012, pp. 697–700.
- [10] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *International conference on machine learning*. PMLR, 2023, pp. 28 492–28 518.